Lab 5 — React Polymorphism & Niche Product CRUD

Multiple Choice Test: React Development

Please select the best answer for each question.

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1. What is the primary purpose of `main.jsx` in a Vite-React project?

A. To define the backend API routes.

B. To act as the entry point connecting React to the HTML `root` div.

C. To compile Java code into JavaScript.

D. To manage the CSS for the entire application.

2. Which hook is used to manage "Component Memory" (data that changes over time)?

A. `useEffect`

B. `useContext`

C. `useState`

D. `useRouter`

3. In JSX, which attribute is used instead of the standard HTML `class`?

A. `classID`

B. `styleClass`

C. `className`

D. `reactClass`

4. What does the "Virtual DOM" allow React to do?

A. Run JavaScript directly in the JVM.

B. Update only the changed parts of the UI rather than the whole page.

C. Connect to an H2 database without a backend.

D. Create physical files on the hard drive during `npm run dev`.

5. When using `fetch()`, what is the purpose of `response.json()`?

A. To save the data to a local `.json` file.

B. To convert the raw HTTP response body into a JavaScript object.

C. To encrypt the data before sending it to Spring Boot.

D. To validate that the CSS is correct.

6. Which command is used to start the local Vite development server?

A. `npm run build`

B. `npm run dev`

C. `mvn spring-boot:run`

D. `git push origin main`

7. What is the "Dependency Array" in a `useEffect` hook?

A. A list of CSS files to import.

B. An array that determines when the effect should re-run.

C. A list of Maven dependencies in the `pom.xml`.

D. A list of all components in the project.

8. If you pass an empty array `[]` as the second argument to `useEffect`, when does the code run?

A. Every time the component re-renders.

B. Never.

C. Only once, when the component is first mounted.

D. Only when the backend server restarts.

9. In React, data is passed from Parent to Child via:

A. State

B. Hooks

C. Props

D. Controllers

10. What happens if you try to modify a prop directly inside a child component?

A. The parent state updates automatically.

B. Props are read-only; request parent-state changes through a callback instead.

C. The browser refreshes.

D. The backend database deletes the record.

11. What is the "Spread Operator" (`...`) typically used for in React state updates?

A. To delete all items in an array.

B. To create a copy of an existing object/array while adding or changing fields.

C. To multiply the price of a book.

D. To import multiple components at once.

12. Why do we use a "Proxy" in `vite.config.js`?

A. To make the website load faster.

B. To bypass CORS issues by forwarding `/api` requests to port 8080.

C. To hide our IP address from the server.

D. To allow multiple users to edit the code at once.

13. What is a "Controlled Component" in React?

A. A component that cannot be deleted.

B. A form input whose value is driven by React state.

C. A component that only works on Windows.

D. A backend controller in Spring Boot.

14. Which JavaScript method is most commonly used to transform an array of data into a list of JSX elements?

A. `.filter()`

B. `.reduce()`

C. `.map()`

D. `.forEach()`

15. Why must every item in a list rendered by `.map()` have a unique `key` prop?

A. To style the items differently.

B. To help React identify which items have changed, been added, or removed.

C. To link the item to a specific CSS file.

D. Because the H2 database requires it for indexing.

16. What is the primary role of the `react-router` library?

A. To speed up the database connection.

B. To handle navigation and URL changes without refreshing the page.

C. To animate buttons.

D. To generate random data for the Seeder.

17. Which component from React Router is used to change the URL instead of an `<a>` tag?

A. `<Route>`

B. `<BrowserRouter>`

C. `<Link>`

D. `<Navbar>`

18. When adapting the guided Magazine CRUD pattern to your niche product, what should determine the form fields and JSON property names?

A. The Magazine field names, unchanged for every product.

B. Your existing niche entity/controller contract, including shared and unique fields and their types.

C. Whatever labels look best, regardless of the backend.

D. Only the CSS class names.

19. After a successful PUT, which check best shows that a niche record was actually saved?

A. Change the displayed text without contacting the backend.

B. Re-fetch the record or reload from the API and confirm the changed shared and unique values.

C. Check that the editor has no spelling warnings.

D. Rename the component file.

20. What is the purpose of `e.preventDefault()` in a form submit handler?

A. To prevent the user from typing.

B. To stop the browser from performing a full page reload.

C. To delete the data before it is sent.

D. To trigger the `useEffect` hook.

21. In React, "State" is:

A. Persistent data stored in the database.

B. Data that stays in memory only until the browser is refreshed.

C. A list of all Java classes.

D. The folder where images are stored.

22. What is a "Fragment" (`<> ... </>`) used for?

A. To break the code into smaller pieces.

B. To wrap multiple JSX elements without adding an extra node to the DOM.

C. To create a background image.

D. To handle 404 errors.

23. Which HTTP method is used by the frontend to create a new record in the backend?

A. GET

B. DELETE

C. POST

D. OPTIONS

24. Which HTTP method is used by the frontend to update an existing record?

A. POST

B. PUT

C. FETCH

D. UPDATE

25. What is the result of `npm run build`?

A. It starts the Spring Boot server.

B. It creates a `dist` (or `static`) folder with optimized production files.

C. It deletes the `node_modules` folder.

D. It generates new random books in the database.

26. How do you pass a function from a Parent to a Child component?

A. You can't pass functions, only strings.

B. Via Props, just like any other variable.

C. Using the `export` keyword.

D. By saving it to the database first.

27. If a component is "Unmounting," it means:

A. It is being saved to the database.

B. It is being removed from the DOM/screen.

C. It is being updated with new props.

D. The user is clicking a button.

28. Which of the following is a "Rule of Hooks"?

A. Hooks can only be used in Java files.

B. Hooks must be called at the top level of a function (not inside an `if` or a loop).

C. Hooks must be named using `UPPER_CASE`.

D. Only one hook is allowed per component.

29. What is "Declarative Programming" in React?

A. Telling the computer exactly how to change every pixel.

B. Describing what the UI should look like for a given state.

C. Declaring all variables as `final`.

D. Writing code only in the `main.jsx` file.

30. In the command `const [count, setCount] = useState(0)`, what is `setCount`?

A. The current value of the state.

B. A function used to update the state and trigger a re-render.

C. A link to the backend controller.

D. An event listener.

31. What is the purpose of the `alt` attribute in an `<img>` tag within JSX?

A. To provide a fallback title.

B. To provide a text description for accessibility and screen readers.

C. To change the image size.

D. To link the image to a book ID.

32. How do you handle a click event on a button in JSX?

A. `onclick="handleClick()"`

B. `onClick={handleClick}`

C. `click={handleClick()}`

D. `on-press={handleClick}`

33. When using React Router, what prop does the `<Route>` component use to match the URL?

A. `link`

B. `to`

C. `path`

D. `href`

34. What is the default port for a Vite development server?

A. 8080

B. 3000

C. 5173

D. 80

35. What is the purpose of the `package.json` file?

A. To store the application's data.

B. To list the project's JavaScript dependencies and scripts.

C. To configure the H2 database.

D. To define the Java package structure.

36. In a standard React CRUD app, where should the "List" state usually live?

A. In the individual Item component.

B. In the common Parent component of the List and the Form.

C. In the `index.html` file.

D. In the CSS file.

37. What happens if you forget to include the `Content-Type: application/json` header in a POST request?

A. The browser will crash.

B. The Spring Boot backend might not be able to parse the JSON body.

C. The price of the book will default to zero.

D. The CSS will stop working.

38. Which hook name is NOT supplied by React or React Router and would need to be implemented as a custom hook in this project?

A. `useBook`

B. `useState`

C. `useEffect`

D. `useParams`

39. To display a JavaScript variable `title` inside JSX, you use:

A. `(title)`

B. `<title>`

C. `{title}`

D. `[title]`

40. What is "Hot Module Replacement" (HMR)?

A. A way to cool down your CPU while coding.

B. Replacing only the changed code in the browser without a full refresh.

C. Swapping out the H2 database for MySQL.

D. A git command to merge branches.

41. Why is it important to use "Immutable" updates (like the spread operator) for state?

A. Because it saves memory.

B. A new object/array reference lets React observe the updated state rather than a mutation of the existing value.

C. Because the backend doesn't accept old objects.

D. Because JSX doesn't support the `push()` method.

42. What component acts as a "Catch-all" for undefined routes in React Router?

A. `<Route path="/" />`

B. `<Route path="*" />`

C. `<NotFound />`

D. `<Default />`

43. In `fetch('/api/rest/books')`, why is the leading slash `/` important?

A. It indicates a folder on the hard drive.

B. It makes the path absolute from the domain root.

C. It tells React to use the Java controller.

D. It is purely optional.

44. Which tool in our stack is responsible for converting JSX into standard JavaScript?

A. Maven

B. Vite with the React plugin

C. Tomcat

D. Git

45. If `loading` is a boolean state, what does `{loading && <Spinner />}` do?

A. It shows the spinner if loading is false.

B. It shows the spinner only if loading is true.

C. It crashes if loading is null.

D. It always shows the spinner.

46. Your frontend request fails because it uses port 3000, but the class backend runs on 8080. What is the best correction?

A. Change the Spring Boot port to 3000.

B. Use the correct relative /api/rest endpoint through the Vite proxy, configured for backend port 8080.

C. Delete the project and start over.

D. Use a different browser.

47. How do you access a specific parameter (like an ID) from the URL in a React component?

A. `props.id`

B. `useState('id')`

C. `useParams()`

D. `window.getURL()`

48. What is the purpose of `parseFloat()` when handling a price input?

A. To make the number look prettier.

B. To ensure a string from an input field is converted to a numeric decimal for the backend.

C. To round the number to the nearest integer.

D. To add a currency symbol.

49. What is the "Entry Point" component that usually holds the Routing logic?

A. `NavBar.jsx`

B. `main.jsx`

C. `App.jsx`

D. `Home.jsx`

50. In a "One-Way Data Flow" architecture:

A. Data goes up, Events go down.

B. Data goes down (via props), Events go up (via callbacks).

C. Data flows directly from the Database to the Component.

D. Components can change each other's state directly.

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1. B	11. B	21. B	31. B	41. B
2. C	12. B	22. B	32. B	42. B
3. C	13. B	23. C	33. C	43. B
4. B	14. C	24. B	34. C	44. B
5. B	15. B	25. B	35. B	45. B
6. B	16. B	26. B	36. B	46. B
7. B	17. C	27. B	37. B	47. C
8. C	18. B	28. B	38. A	48. B
9. C	19. B	29. B	39. C	49. C
10. B	20. B	30. B	40. B	50. B